
RAPID AND AUTOMATED ALLOY DESIGN WITH GRAPH NEURAL NETWORK-POWERED LLM-DRIVEN MULTI-AGENT SYSTEMS

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Abstract

A multi-agent AI model is used to automate the discovery of new metallic alloys, integrating multimodal data and external knowledge including insights from physics via atomistic simulations. Our multi-agent system features three key components: (a) a suite of LLMs responsible for tasks such as reasoning and planning, (b) a group of AI agents with distinct roles and expertise that dynamically collaborate, and (c) a newly developed graph neural network (GNN) model for rapid retrieval of key physical properties. A set of LLM-driven AI agents collaborate to automate the exploration of the vast design space of MPEAs, guided by predictions from the GNN. We focus on the NbMoTa family of body-centered cubic (bcc) alloys, modeled using an ML-based interatomic potential, and target two key properties: the Peierls barrier and solute/screw dislocation interaction energy. Our GNN model accurately predicts these atomic-scale properties, providing a faster alternative to costly brute-force calculations and reducing the computational burden on multi-agent systems for physics retrieval. This AI system revolutionizes materials discovery by reducing reliance on human expertise and overcoming the limitations of direct all-atom simulations. By synergizing the predictive power of GNNs with the dynamic collaboration of LLM-based agents, the system autonomously navigates vast alloy design spaces, identifying trends in atomic-scale material properties and predicting macro-scale mechanical strength, as demonstrated by several computational experiments. This approach accelerates the discovery of advanced alloys and holds promise for broader applications in other complex systems, marking a significant step forward in automated materials design.

Keywords Multi-agent system, Large language model · Deep learning · Graph neural networks · complex alloys · Materials design · Scientific machine learning

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